

# PRO5-A1140-OSB

Serie PRO Microchannel · R454B · senza serbatoio / without tank



PRO5-A1140-OSB

RESA FRIGORIFERA · COOLING CAPACITY

**754** kW

EER · 12/7 °C · AIR 35 °C

**2,88**

RESA NOMINALE · 15 °C · AIR 25 °C

**1.133** kW

PORTATA ACQUA · WATER FLOW

**129,7** m<sup>3</sup>/h

|                            |                 |                                        |                       |
|----------------------------|-----------------|----------------------------------------|-----------------------|
| Compressori / Compressors  | 6 × scroll      | L × P × H (mm)                         | 8.120 × 2.197 × 2.442 |
| Circuiti / Circuits        | 2               | Serbatoio / Tank                       | —                     |
| Ventilatori / Fans         | 14              | Taglia / Size                          | 1140                  |
| Refrigerante / Refrigerant | R454B · GWP 466 | Campo aria esterna / Ambient air range | -5 / +47 °C           |

## Resa in sintesi / Capacity at a glance <sup>kW</sup>

| ARIA ESTERNA / AMBIENT AIR                  | 25 °C | 30 °C | 35 °C | 40 °C | 45 °C |
|---------------------------------------------|-------|-------|-------|-------|-------|
| Acqua in uscita 7 °C / Leaving water 7 °C   | 859   | 808   | 754   | 700   | 640   |
| Acqua in uscita 15 °C / Leaving water 15 °C | 1.133 | 1.067 | 999   | 926   | 854   |

Refrigeratori d'acqua condensati ad aria per processo industriale e sistemi HVAC, con compressori scroll in tandem che entrano in funzione in sequenza per un'efficienza elevata anche ai carichi parziali.

Air-cooled water chillers for industrial processes and HVAC systems, with tandem scroll compressors staged in sequence for high efficiency at part load.

# Dati tecnici

## Technical data

PR05-A1140-0SB

**Prestazioni nominali — acqua in uscita 15 °C, aria 25 °C / Nominal performance — 15 °C leaving water, 25 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 1.133,22 kW              |
| Resa frigorifera / Cooling capacity | 974.569 kcal/h           |
| Potenza assorbita / Power input     | 206,49 kW                |
| EER / EER                           | 5,49                     |
| Portata acqua / Water flow          | 194,91 m <sup>3</sup> /h |

**Condizioni standard — acqua 12/7 °C, aria 35 °C / Standard conditions — 12/7 °C water, 35 °C air**

|                                     |                          |
|-------------------------------------|--------------------------|
| Resa frigorifera / Cooling capacity | 754,23 kW                |
| Potenza assorbita / Power input     | 262,10 kW                |
| EER / EER                           | 2,88                     |
| ESEER / ESEER                       | 4,13                     |
| Portata acqua / Water flow          | 129,73 m <sup>3</sup> /h |

### Componenti / Components

|                                             |                       |
|---------------------------------------------|-----------------------|
| Compressori / Compressors                   | 6 × Scroll            |
| Circuiti frigoriferi / Refrigerant circuits | 2                     |
| Ventilatori / Fans                          | 14                    |
| Condensatore / Condenser                    | a microcanali         |
| Evaporatore / Evaporator                    | a fascio tubiero      |
| Refrigerante / Refrigerant                  | R454B · GWP 466 · A2L |

### Installazione / Installation

|                                                                                           |                           |
|-------------------------------------------------------------------------------------------|---------------------------|
| Dimensioni L × P × H / Dimensions L × D × H                                               | 8.120 × 2.197 × 2.442 mm  |
| Campo aria esterna / Ambient air range                                                    | -5 / +47 °C               |
| Dati elettrici, carica gas, peso, attacchi / Electrical data, charge, weight, connections | su richiesta / on request |

### Dotazioni di serie / Standard equipment

- Valvole di intercettazione  
Shut-off valves
- Griglia di protezione condensatore  
Condenser protection grille
- Valvola di espansione elettronica  
Electronic expansion valve
- Condensatore a microcanali  
Micro-channel condenser
- Evaporatore a fascio tubiero  
Shell & tube evaporator
- Display 4"/7"  
4"/7" display
- Pressostati e manometri  
Pressure switches and gauges
- Pressostato differenziale acqua  
Water differential pressure switch
- Presa di servizio 220 V e luce quadro elettrico  
220 V service socket and electrical panel light

### Con glicole / With glycol

| MISCELA  | COEFF.  | 12/7 · 35 °C |
|----------|---------|--------------|
| MEG 20 % | × 0,980 | 739,1 kW     |
| MEG 30 % | × 0,950 | 716,5 kW     |
| MEG 40 % | × 0,920 | 693,9 kW     |
| MPG 20 % | × 0,975 | 735,4 kW     |
| MPG 30 % | × 0,960 | 724,1 kW     |
| MPG 40 % | × 0,950 | 716,5 kW     |

# Prestazioni Performance

PRO5-A1140-OSB · acqua pura,  $\Delta T$  5 K · pure water, 5 K  $\Delta T$  ·  $\square$  12/7 °C · 35 °C

## Resa frigorifera / Cooling capacity (kW)

| LWT ↓ · air → | 15    | 20    | 25    | 30    | 35  | 40  | 45  |
|---------------|-------|-------|-------|-------|-----|-----|-----|
| 1             | 770   | 732   | 692   | 648   | 604 | 559 | 511 |
| 2             | 799   | 760   | 718   | 673   | 628 | 580 | 532 |
| 3             | 828   | 787   | 744   | 697   | 652 | 601 | 552 |
| 4             | 859   | 817   | 772   | 724   | 677 | 625 | 574 |
| 5             | 890   | 846   | 800   | 751   | 702 | 649 | 597 |
| 6             | 923   | 877   | 830   | 780   | 728 | 674 | 618 |
| 7             | 955   | 908   | 859   | 808   | 754 | 700 | 640 |
| 8             | 990   | 941   | 891   | 839   | 783 | 724 | 665 |
| 9             | 1.025 | 974   | 922   | 869   | 811 | 749 | 691 |
| 10            | 1.061 | 1.010 | 956   | 899   | 840 | 778 | 717 |
| 11            | 1.096 | 1.045 | 989   | 930   | 869 | 807 | 743 |
| 12            | 1.135 | 1.081 | 1.024 | 964   | 900 | 836 | 770 |
| 13            | 1.173 | 1.118 | 1.058 | 997   | 931 | 865 | 796 |
| 14            | 1.214 | 1.156 | 1.096 | 1.032 | 965 | 895 | 825 |
| 15            | 1.254 | 1.194 | 1.133 | 1.067 | 999 | 926 | 854 |

## Potenza assorbita / Power input (kW)

| LWT ↓ · air → | 15  | 20  | 25  | 30  | 35  | 40  | 45  |
|---------------|-----|-----|-----|-----|-----|-----|-----|
| 1             | 171 | 191 | 212 | 235 | 261 | 290 | 324 |
| 2             | 171 | 191 | 212 | 235 | 261 | 291 | 324 |
| 3             | 170 | 190 | 212 | 235 | 261 | 291 | 325 |
| 4             | 170 | 190 | 212 | 235 | 261 | 291 | 325 |
| 5             | 169 | 190 | 212 | 235 | 262 | 291 | 325 |
| 6             | 169 | 190 | 212 | 236 | 262 | 292 | 325 |
| 7             | 168 | 189 | 211 | 236 | 262 | 292 | 325 |
| 8             | 167 | 189 | 211 | 236 | 262 | 292 | 325 |
| 9             | 166 | 188 | 211 | 235 | 262 | 292 | 326 |
| 10            | 164 | 187 | 210 | 235 | 262 | 292 | 326 |
| 11            | 163 | 186 | 210 | 235 | 262 | 292 | 326 |
| 12            | 162 | 185 | 209 | 235 | 262 | 292 | 326 |
| 13            | 160 | 184 | 208 | 234 | 262 | 292 | 326 |
| 14            | 158 | 183 | 207 | 234 | 262 | 292 | 326 |
| 15            | 156 | 181 | 206 | 233 | 261 | 292 | 326 |

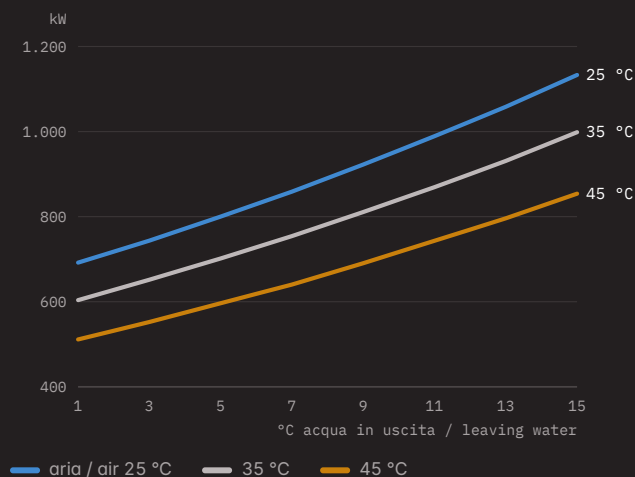
## EER / EER (kW/kW)

| LWT ↓ · air → | 15   | 20   | 25   | 30   | 35   | 40   | 45   |
|---------------|------|------|------|------|------|------|------|
| 1             | 4,50 | 3,84 | 3,27 | 2,76 | 2,32 | 1,92 | 1,58 |
| 2             | 4,68 | 3,99 | 3,39 | 2,86 | 2,41 | 2,00 | 1,64 |
| 3             | 4,86 | 4,13 | 3,51 | 2,97 | 2,49 | 2,07 | 1,70 |
| 4             | 5,06 | 4,29 | 3,64 | 3,08 | 2,59 | 2,15 | 1,77 |
| 5             | 5,26 | 4,45 | 3,78 | 3,19 | 2,68 | 2,23 | 1,84 |
| 6             | 5,47 | 4,63 | 3,92 | 3,31 | 2,78 | 2,31 | 1,90 |
| 7             | 5,69 | 4,80 | 4,06 | 3,43 | 2,88 | 2,40 | 1,97 |
| 8             | 5,94 | 4,99 | 4,22 | 3,56 | 2,98 | 2,48 | 2,04 |
| 9             | 6,18 | 5,19 | 4,37 | 3,69 | 3,09 | 2,57 | 2,12 |
| 10            | 6,45 | 5,40 | 4,54 | 3,82 | 3,20 | 2,66 | 2,20 |
| 11            | 6,72 | 5,61 | 4,71 | 3,96 | 3,31 | 2,76 | 2,28 |
| 12            | 7,02 | 5,84 | 4,89 | 4,11 | 3,43 | 2,86 | 2,36 |
| 13            | 7,33 | 6,08 | 5,08 | 4,26 | 3,55 | 2,96 | 2,44 |
| 14            | 7,68 | 6,33 | 5,28 | 4,42 | 3,69 | 3,06 | 2,53 |
| 15            | 8,03 | 6,59 | 5,49 | 4,58 | 3,82 | 3,17 | 2,62 |

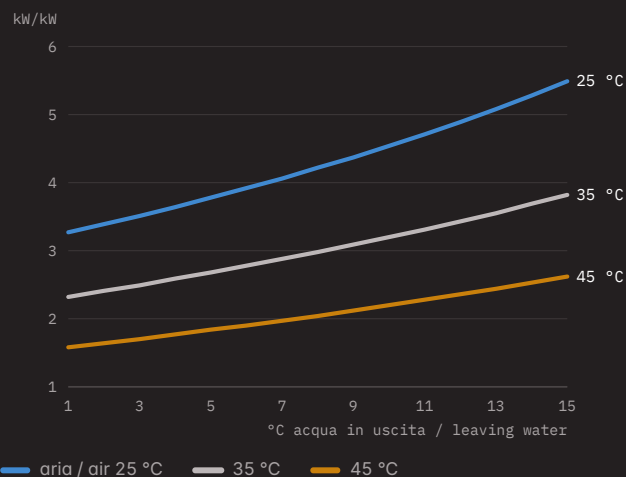
# Curve e dimensioni

## Curves and dimensions

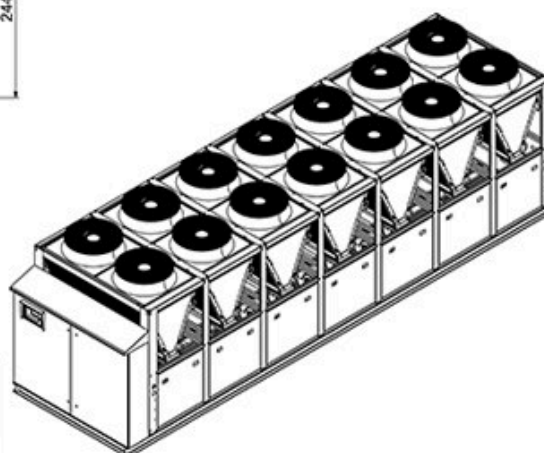
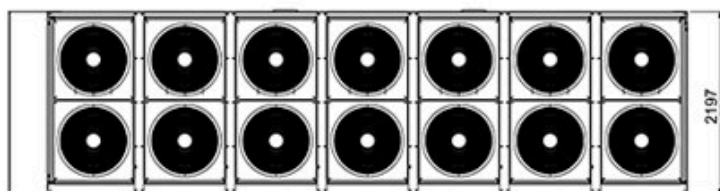
### Resa frigorifera / Cooling capacity



### EER / EER



### Dimensioni / Dimensions · L × P × H 8.120 × 2.197 × 2.442 mm



Disegno indicativo, quote in mm (±10 mm). Dimensioni da confermare in fase d'ordine. / Indicative drawing, dimensions in mm (±10 mm), to be confirmed at order stage.

# Optional — 1/2 Options

PRO5-A1140-OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

## Pompe / Pumps

|                                                                                                     |
|-----------------------------------------------------------------------------------------------------|
| Pompa 132 m³/h – prevalenza 25 m<br>Pump 132 m³/h – head 25 m                                       |
| Pompa 132 m³/h – prevalenza 40 m<br>Pump 132 m³/h – head 40 m                                       |
| Pompa doppia montata 132 m³/h – prevalenza 25 m<br>Twin pump, factory -fitted, 132 m³/h – head 25 m |
| Pompa doppia montata 132 m³/h – prevalenza 40 m<br>Twin pump, factory -fitted, 132 m³/h – head 40 m |

## Serbatoi e vasi di espansione / Tanks and expansion vessels

|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory -fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory -fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory -fitted twin AISI 304 pump    |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa doppia montata<br>External water tank 2000 l, AISI 304 with factory -fitted twin pump                         |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa doppia montata<br>External water tank 3000 l, AISI 304 with factory -fitted twin pump                         |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa doppia montata<br>External water tank 5000 l, AISI 304 with factory -fitted twin pump                         |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 1500 l, AISI 304 with factory -fitted single AISI 304 pump |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 2000 l, AISI 304 with factory -fitted single AISI 304 pump |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 3000 l, AISI 304 with factory -fitted single AISI 304 pump |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola in AISI 304 montata<br>External water tank 5000 l, AISI 304 with factory -fitted single AISI 304 pump |
| Serbatoio idraulico esterno 1500 l in AISI 304 con pompa singola montata<br>External water tank 1500 l, AISI 304 with factory -fitted single pump                      |
| Serbatoio idraulico esterno 2000 l in AISI 304 con pompa singola montata<br>External water tank 2000 l, AISI 304 with factory -fitted single pump                      |
| Serbatoio idraulico esterno 3000 l in AISI 304 con pompa singola montata<br>External water tank 3000 l, AISI 304 with factory -fitted single pump                      |
| Serbatoio idraulico esterno 5000 l in AISI 304 con pompa singola montata<br>External water tank 5000 l, AISI 304 with factory -fitted single pump                      |

|                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Serbatoio idraulico esterno 2000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 2000 l, pressurised, galvanised steel with factory -fitted twin pump    |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 3000 l, pressurised, galvanised steel with factory -fitted twin pump    |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa doppia montata<br>External water tank 5000 l, pressurised, galvanised steel with factory -fitted twin pump    |
| Serbatoio idraulico esterno 1500 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 1500 l, pressurised, galvanised steel with factory -fitted single pump |
| Serbatoio idraulico esterno 3000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 3000 l, pressurised, galvanised steel with factory -fitted single pump |
| Serbatoio idraulico esterno 5000 l pressurizzato in acciaio zincato con pompa singola montata<br>External water tank 5000 l, pressurised, galvanised steel with factory -fitted single pump |
| Vaso di espansione 100 l<br>Expansion vessel 100 l                                                                                                                                          |
| Vaso di espansione 200 l<br>Expansion vessel 200 l                                                                                                                                          |
| Vaso di espansione 300 l<br>Expansion vessel 300 l                                                                                                                                          |
| Vaso di espansione 500 l<br>Expansion vessel 500 l                                                                                                                                          |
| Vaso di espansione 750 l<br>Expansion vessel 750 l                                                                                                                                          |

## Valvole e filtri / Valves and filters

|                                                       |
|-------------------------------------------------------|
| Elettrovalvola 2"<br>Solenoid valve 2"                |
| Elettrovalvola 3"<br>Solenoid valve 3"                |
| Elettrovalvola 4"<br>Solenoid valve 4"                |
| Elettrovalvola 5"<br>Solenoid valve 5"                |
| Elettrovalvola 6"<br>Solenoid valve 6"                |
| Filtro a dischi in acciaio 2"<br>Steel disc filter 2" |

# Optional — 2/2

## Options

PRO5-A1140-OSB · 65 optional disponibili per questa configurazione · 65 options available for this configuration

### Valvole e filtri / Valves and filters

Filtro a dischi in acciaio 3"

Steel disc filter 3"

Filtro a dischi in acciaio 4"

Steel disc filter 4"

Filtro meccanico a Y 2"

Y-strainer 2"

Filtro meccanico a Y 3"

Y-strainer 3"

Filtro meccanico a Y 4"

Y-strainer 4"

Filtro meccanico a Y 5"

Y-strainer 5"

Filtro meccanico a Y 6"

Y-strainer 6"

Valvola a 3 vie 2" miscelatrice

3-way mixing valve 2"

Valvola a 3 vie 3" miscelatrice

3-way mixing valve 3"

Valvola a 3 vie 4" miscelatrice

3-way mixing valve 4"

Valvola a 3 vie 5" miscelatrice

3-way mixing valve 5"

Valvola a 3 vie 6" miscelatrice

3-way mixing valve 6"

Valvola a 3 vie 2" on/off

3-way valve 2" on/off

Valvola a 3 vie 3" on/off

3-way valve 3" on/off

Valvola a 3 vie 4" on/off

3-way valve 4" on/off

Valvola a 3 vie 5" on/off

3-way valve 5" on/off

Valvola a 3 vie 6" on/off

3-way valve 6" on/off

### Circuito frigorifero / Refrigerant circuit

Cofanatura insonorizzante compressori

Compressor acoustic enclosure

Compressore inverter

Inverter compressor

Desurriscaldatore a piastre

Plate desuperheater

Rilevatore fughe gas

Refrigerant leak detector

### Ventilazione e clima / Fans and climate

Inverter ventilatori

Fan inverter

Ventilatori EC

EC fans

### Controllo e comunicazione / Control and communication

Comando remoto con display (cavo escluso)

Remote control with display (cable not included)

Comunicazione RS485 Modbus RTU / TCP-IP (cavo escluso)

RS485 Modbus RTU / TCP-IP communication (cable not included)

Comunicazione RS485 verso BACnet (cavo escluso)

RS485 to BACnet communication (cable not included)

### Quadro elettrico / Electrical panel

Analizzatore di rete

Power analyser

Kit invernale (inverter ventilatori + resistenza quadro + resistenza

evaporatore + termostato) fino a -25 °C

Winter kit (fan inverter + panel heater + evaporator heater + thermostat) down to -25 °C

### Installazione / Installation

Antivibranti a molla

Spring anti-vibration mounts

Antivibranti in gomma

Rubber anti-vibration mounts

## Note Notes

- Prestazioni con acqua pura, salto termico 5 K. Le condizioni di riferimento sono indicate in ogni tabella.
  - Con miscele di glicole la resa frigorifera si corregge con i coefficienti della tabella dedicata; potenza assorbita ed EER restano invariati.
  - Dimensioni ricavate dai disegni quotati (tolleranza  $\pm 10$  mm): da confermare in fase d'ordine.
  - Dati elettrici, carica di refrigerante, pesi e attacchi idraulici sono forniti su richiesta per la configurazione scelta.
  - Procryon si riserva di modificare dati e caratteristiche senza preavviso, per il miglioramento continuo del prodotto.
- Performance with pure water, 5 K temperature difference. Reference conditions are stated in each table.
  - With glycol mixtures, cooling capacity is corrected with the coefficients in the dedicated table; power input and EER are unchanged.
  - Dimensions taken from dimensional drawings (tolerance  $\pm 10$  mm): to be confirmed at order stage.
  - Electrical data, refrigerant charge, weights and water connections are available on request for the chosen configuration.
  - Procryon reserves the right to change data and features without notice, as part of continuous product improvement.

## Parliamoci Let's talk

Progettazione, intervento urgente, valutazione di un impianto esistente, supporto tecnico e intervento rapido — 24/7.

System design, urgent response, assessment of an existing plant, technical support and rapid response — 24/7.

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Continuity by design